

ACTIVE STABILITY SYSTEM FOR HIGH-SPEED DRIVING

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Abstract

Modern autonomous racing requires control architectures that navigate the narrow boundary between maximum velocity and dynamic instability. Conventional path-tracking algorithms, like Pure Pursuit and Stanley, rely on kinematic assumptions neglecting vehicle mass, yaw inertia, and tire-road friction. Consequently, these systems suffer catastrophic spin-out failures when lateral accelerations approach Kamm Friction Circle limits. This study presents a proactive Active Stability System developed through a rigorous 100% Model-Based Design (MBD) paradigm. Utilizing a high-fidelity, non-linear 3-Degree-of-Freedom (3DOF) Single Track plant, the research evaluates five evolutionary control models, culminating in an Explicit Model Predictive Controller (MPC) integrated with a Smart Throttle longitudinal governor. Unlike reactive safety systems that intervene via momentum-reducing braking, the MPC proactively optimizes steering and longitudinal forces to maintain stability at physical tire slip saturation limits. Stress-tested against significant sensor latency and Gaussian noise, experimental results demonstrate the MPC consistently achieves the lowest cross-track Root Mean Square Error (RMSE) against all baselines. Furthermore, the MPC uniquely executed high-speed obstacle avoidance without compromising dynamic equilibrium. This proves that a unified predictive framework accounting for tire saturation and actuator constraints is essential for next-generation autonomous mobility.

Keywords: Active Stability System / High-Speed Autonomous Driving / Model-Based Design
/ Model Predictive Control (MPC) / Vehicle Dynamics

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TABLE OF CONTENTS

	PAGE
ABSTRACT	ii
ACKNOWLEDGEMENTS	iii
TABLE OF CONTENTS	iv
LIST OF TABLES	vi
LIST OF FIGURES	vii
LIST OF SYMBOLS	viii
LIST OF TECHNICAL VOCABULARY AND ABBREVIATIONS	xi
 CHAPTER	
1. INTRODUCTION	1
1.1 Background and Motivation	1
1.2 Existing Solutions and Their Limitations	1
1.3 Problem Statement	2
1.4 Project Objectives	3
1.5 Scope of the Study and the MBD Pivot	4
1.6 Expected Benefits	5
 2. SYSTEM MODELLING	6
2.1 The 3DOF Vehicle Dynamics Plant	6
2.2 System Inputs and Outputs Architecture	8
2.3 Tire Dynamics and the Slip Boundary	9
2.4 Environmental Modeling: Track Generation	10
 3. LATERAL TRACKING CONTROLLERS	11
3.1 Baseline Longitudinal Control Law (Models 1 to 4)	11
3.2 Model 1: Pure Pursuit with Dynamic Lookahead	12
3.3 Model 2: The Stanley Controller	14
3.4 Model 3: Classical Active Yaw Control	16

CONTENTS (Cont'd)

	PAGE
3.5 Model 4: Optimal State-Space Control (LQR)	19
3.6 Model 5: Predictive Optimization & Obstacle Avoidance (MPC + APF)	21
3.7 Simulink Implementation Details	26
 4. SIMULATION RESULTS & COMPARATIVE ANALYTICS	 27
4.1 Simulation Environmental Setup	27
4.2 Performance Metrics	28
4.3 Comprehensive Performance Synthesis	29
4.4 Comparative Graph Analysis	29
 5. DISCUSSION, CONCLUSION, AND SUGGESTION	 35
5.1 Discussion of Simulation Results	35
5.2 Conclusion	37
5.3 Engineering Professionalism and Teamwork	38
5.4 Suggestions for Future Work	40
 REFERENCES	 41
 APPENDIX A	 42
A.1 Repository Architecture	43
A.2 Simulation Execution & Reproducibility Guide	43
A.3 Digital Twin Media & Simulation Videos	44
A.4 Authors	45

LIST OF TABLES

TABLE	PAGE
2.0 Physical 3DOF Plant Parameters	8
4.0 Final Benchmark Results (Target Speed: 20.0 m/s)	29
5.0 Master Simulation Performance Summary (20.0 m/s Benchmark)	35

LIST OF FIGURES

FIGURE	PAGE
1.1 The "Kinematic Trap" Feedback Loop in Geometric Path Planning	3
1.2 The Model-Based Design (MBD) V-Model Workflow	5
2.1 Free Body Diagram of the 3DOF Single Track Model	6
2.2 System Inputs/Outputs Block Diagram	8
2.3 1:10 Scaled Raceline of the Austin Track	10
3.1 Generalized Closed-Loop MBD Architecture Across All Models	11
3.2 Pure Pursuit Geometric Lookahead Vectors	13
3.3 Comparative kinematic tracking logic: Pure Pursuit vs. Stanley	15
3.4 Control loop schematic of the Model 3 Hybrid Stanley with Active Yaw	16
3.5 Control loop block diagram of the Model 5 Unified Predictive MPC	22
4.1 Simulink Master Simulation Environment and Scope Setup	27
4.2 Trajectory Overlay: LQR vs PurePursuit	30
4.3 LQR vs MPC Performance (Velocity and CTE Comparison)	31
4.4 Model 5 Track Telemetry & Trajectory	32
4.5 Global map trajectory tracking telemetry showing Model 5 multi-obstacle avoidance	33
5.1 Phase Portrait (Slip Angle β vs Yaw Rate r)	37

LIST OF SYMBOLS

SYMBOL		UNIT
A, B	Continuous-time system dynamics matrix and input matrix	-
a	Longitudinal distance from center of mass to front axle	m
a_y	Lateral acceleration of the vehicle chassis	m/s ²
b	Longitudinal distance from center of mass to rear axle	m
$C_{\alpha f}, C_{\alpha r}$	Cornering stiffness coefficient of front and rear tires	N/rad
d	Euclidean distance from vehicle Center of Gravity to obstacle	m
d_0	Artificial Potential Field sensor range of influence	m
e_{apf}	Lateral reference trajectory offset injected by repulsion field	m
e_{fa}	Lateral cross-track error measured relative to the front axle	m
e_r	Yaw rate regulation tracking deviation error	rad/s
e_y	Cross-track error calculated relative to the vehicle reference segment	m
F_{rep}	Repulsive potential force vector acting on the predictive model	N
F_x	Longitudinal propulsion force commanded by the controller	N
F_{xf}, F_{xr}	Longitudinal tire forces acting on the front and rear axles	N
F_{yf}, F_{yr}	Lateral tire forces acting on the front and rear axles	N
g	Acceleration due to gravity (9.81)	m/s ²
h	Vertical distance from ground plane to center of mass	m
I_{zz}	Chassis moment of inertia about the vertical yaw axis	kg.m ²
J_{lqr}, J_{mpc}	Quadratic optimization cost objective function index profiles	-
k	Current discrete execution time step	-
K_{lqr}	Optimal state feedback gain tracking vector row matrix	-
K_p, K_i, K_d	Proportional, integral, and derivative control loop gain coefficients	-
k_s	Stanley position non-linear error feedback scaling gain	-
k_{soft}	Stanley softening velocity term to prevent division singularities	m/s

LIST OF SYMBOLS (Cont'd)

SYMBOL		UNIT
K_{us}	Analytical vehicle understeer gradient constant factor	s ² /m
K_v	Optimal feedforward gain parameter for steady-state cornering	-
k_v	Velocity-scaling gain for geometric lookahead search governors	s
L	Total vehicle wheelbase distance from front to rear axle	m
l_d	Spatial lookahead pursuit target search distance matrix boundary	m
$l_{d,min}$	Lower bound lookahead constraint limit threshold	m
m	Total physical mass of the rigid vehicle body platform	kg
m, p	Control and prediction horizon time steps for the predictive loop steps	-
Q, R	State tracking error and control effort penalty weighting matrices	-
r, r_{ref}	Actual and steady-state reference target vehicle yaw rates	rad/s
R_{local}	Instantaneous geometric turning radius of track curvature segments	m
T_s	Fixed execution time step interval sampling configuration	s
U_{rep}	Artificial potential field repulsive energy field equation state	J
v_{max}, v_{ref}	Absolute theoretical maximum reference velocity	m/s
v_{target}	Dynamically governed safe reference target velocity	m/s
v_x, v_y	Longitudinal and lateral spatial vehicle chassis frame velocities	m/s
X, Y	Global spatial coordinates derived via velocity transformation	m
x, u	State variable vector and control input vector	-
x_g, y_g	Global spatial coordinates of the target waypoint on the reference path	m
x_{cg}, y_{cg}	Global spatial coordinates of the vehicle's center of gravity	m
α_f, α_r	Instantaneous lateral tire slip angles at the front and rear axles	rad or °
α_{look}	Geometric heading error angle to the chosen path waypoint	rad
β	Vehicle body side-slip angle relative to the center of mass	rad
δ, δ_{track}	Commanded tire steering angle and baseline path steering angle	rad or °
δ_{ff}	Open-loop feedforward steering angle command	rad or °

LIST OF SYMBOLS (Cont'd)

SYMBOL		UNIT
$\Delta\delta_{yaw}$	Corrective inner stability loop active yaw yaw-rate steering offset	rad or °
η	Repulsive scaling gain factor for structural field potential grids	-
θ, θ_{track}	Vehicle heading orientation angle and local reference path heading	rad
θ_e	Heading alignment orientation error mismatch vector angle	rad
κ	Mathematical curvature line representation of path segments	m ⁻¹
μ	Dimensionless road-tire maximum peak friction coefficient limit	-
τ	Injected computational transport and actuator latency delay limit	s
ψ	Vehicle global yaw angle derived from yaw rate integration	rad

LIST OF TECHNICAL VOCABULARY AND ABBREVIATIONS

3DOF	Three Degrees of Freedom (Longitudinal, Lateral, and Yaw axes)
ABET	Accreditation Board for Engineering and Technology
ADAS	Advanced Driver Assistance Systems
APF	Artificial Potential Fields
ARE	Algebraic Riccati Equation
AUTOSAR	Automotive Open System Architecture
AV	Autonomous Vehicle
BLDC	Brushless Direct Current Motor
CAN	Controller Area Network Bus Communication Protocol
CG	Center of Gravity
COTA	Circuit of the Americas
CTE	Cross-Track Error
ECU	Electronic Control Unit
EKF	Extended Kalman Filter
ENE	Electronics and Instrumentation Engineering Department
ESC	Electronic Stability Control
ESP	Electronic Stability Program (also ESP32 Microcontroller)
HIL	Hardware-in-the-Loop
HUD	Heads-Up Display
I/O	Input/Output
IEEE	Institute of Electrical and Electronics Engineers
IMU	Inertial Measurement Unit
ISO	International Organization for Standardization
KMUTT	King Mongkut's University of Technology Thonburi
LQR	Linear Quadratic Regulator
MBD	Model-Based Design
MEMS	Micro-Electro-Mechanical Systems

LIST OF TECHNICAL VOCABULARY AND ABBREVIATIONS (Cont'd)

MISRA	Motor Industry Software Reliability Association
MPC	Model Predictive Control (and Controller)
PI	Proportional-Integral Controller
PID	Proportional-Integral-Derivative Controller
RMSE	Root Mean Square Error
ROS 2	Robot Operating System 2 Framework Middleware
RWD	Rear-Wheel Drive Vehicle Layout Configuration
SAE	Society of Automotive Engineers
SD	Secure Digital (Memory Card)
SSD	Solid State Drive
VESC	Vedder Electronic Speed Controller Actuator

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

As autonomous driving technology transitions from low-speed urban environments to high-speed highways, active safety systems must evolve. Traditional Advanced Driver Assistance Systems (ADAS), such as the Electronic Stability Program (ESP), operate reactively; they wait for a vehicle to lose traction before applying differential braking forces to stabilize the chassis. While effective for human-driven vehicles, this reactive braking degrades the trajectory tracking accuracy of autonomous path planners and unnecessarily bleeds off vehicle momentum.

The motivation for this project is to develop a proactive control architecture that treats vehicle stability and path tracking as a single, unified optimization problem. By actively controlling both steering and longitudinal propulsion, an autonomous vehicle can maintain stable path tracking even when executing emergency maneuvers at high speeds. This approach ensures safety without sacrificing vehicle speed or trajectory accuracy. The predictive stability control logic developed in this study is highly applicable to high-speed highway commuters, autonomous racing platforms, and advanced motorsport engineering.

1.2 Existing Solutions and Their Limitations

A review of existing automotive path-tracking methods reveals a critical gap between low-speed geometric tracking and high-speed dynamic stability management:

1. **Pure Pursuit (Geometric):** Tracks paths by calculating a steering angle based on a lookahead point on the path. This method is computationally simple but does not account for vehicle lateral slip, causing severe tracking drift at speeds above 5 m/s.
2. **Stanley Controller (Geometric):** Uses a non-linear control law to minimize cross-track error at the front axle. Although more accurate than Pure Pursuit, Stanley commands aggressive steering angles during large deviations, saturating the front tires and causing severe yaw instability at high speeds.

3. **Electronic Stability Program (ESP) with Yaw Control:** Modern consumer vehicles use yaw rate feedback to apply differential braking. While effective at stabilizing the chassis, ESP is purely reactive and slows the vehicle down significantly, which is undesirable for optimal, continuous path tracking.

1.3 Problem Statement

The central problem addressed in this study is the Kinematic Trap. Typical path-tracking controllers (like Pure Pursuit and Stanley) are designed using the Kinematic Bicycle Model, which assumes that the vehicle's tires do not slip laterally:

$$v_y = 0 \implies \beta = 0 \quad (1.1)$$

Where v_y is the lateral velocity and β is the vehicle slip angle. This assumption holds true at low speeds. However, as longitudinal velocity (v_x) increases, the tire forces required to negotiate curves must overcome centrifugal inertia. Under these conditions, the wheels develop a tire slip angle (α), which is the angle between the wheel's orientation and its true velocity vector.

When the lateral force demand on the front tires exceeds the physical road friction coefficient (μ), the front tire slip angle (α_f) passes its linear limit (typically $\alpha_f \approx 5^\circ$). At this point, the tires saturate. Demanding additional steering input does not increase lateral acceleration; instead, it reduces force, trapping geometric controllers in a destabilizing feedback loop that leads to terminal understeer or spin-out. This catastrophic cycle, where increased steering demand actively reduces lateral force capability, is illustrated in **Figure 1.1**.

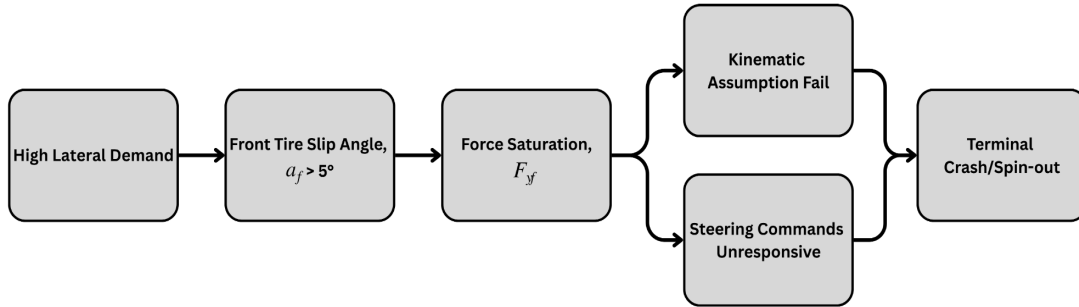


Figure 1.1: The "Kinematic Trap" Feedback Loop in Geometric Path Planning

1.4 Project Objectives

To resolve the Kinematic Trap and bridge the gap between path tracking and stability, this project aims to achieve the following core objectives:

1. **Develop a High-Fidelity Plant:** Model a non-linear 3-Degree-of-Freedom (3DOF) Single Track vehicle dynamics model in Simulink to serve as the "Digital Twin" for deterministic simulation.

2. **Formulate a Stability Metric:** Monitor the front and rear tire slip angles (α_f, α_r) and vehicle slip angle (β) to establish a clear mathematical boundary for stable driving.

3. **Implement an Evolutionary Controller Benchmark:** Design, implement, and compare five control architectures via simulation:

3.1) **Model 1 (Pure Pursuit):** Kinematic, lookahead-based steering baseline.

3.2) **Model 2 (Stanley):** Kinematic, cross-track error-based steering baseline.

3.3) **Model 3 (Active Yaw):** Cascaded proportional-integral-derivative (PID) feedback controller.

3.4) **Model 4 (Linear Quadratic Regulator):** Optimal state-space state feedback controller.

3.5) **Model 5 (MPC + APF):** Explicit Model Predictive Controller with Artificial Potential Fields and a "Smart Throttle" governor.

4. **Demonstrate Obstacle Avoidance:** Successfully navigate static obstacles placed along a racing line at high speeds (20 m/s) while remaining within 0.7 m of the track center-line without inducing a spin-out.

1.5 Scope of the Study and the MBD Pivot

In the early phases of this work (Semester 1), the research focused on physical implementation using a 1:10 scale RC car chassis equipped with an NVIDIA Jetson Nano running ROS 2. However, physical testing revealed significant hardware constraints:

1. **Sensor Noise and Latency:** Low-cost MEMS Inertial Measurement Units (IMUs) suffered from drift, and wheel encoders experienced lag, which corrupted high-frequency state estimation.
2. **Actuator Backlash:** Mechanical tolerance in the steering linkages introduced unmodeled delays.
3. **Safety Constraints:** Inducing high-speed spin-outs at 20 m/s presented an immediate risk of destroying the hardware.

Rather than restricting the research to safe, low-speed maneuvers, the project team made a strategic decision to pivot to a 100% Model-Based Design (MBD) paradigm. By transferring the physical parameters of the 8.0 kg chassis to a simulation model, the team built a validated Digital Twin. This allowed them to analyze tire saturation physics up to 20 m/s in a deterministic environment, ensuring both physical safety and absolute mathematical reproducibility. The structured approach used to transition from hardware constraints to a fully deterministic simulation environment is outlined in the Model-Based Design (MBD) workflow shown in **Figure 1.2**.

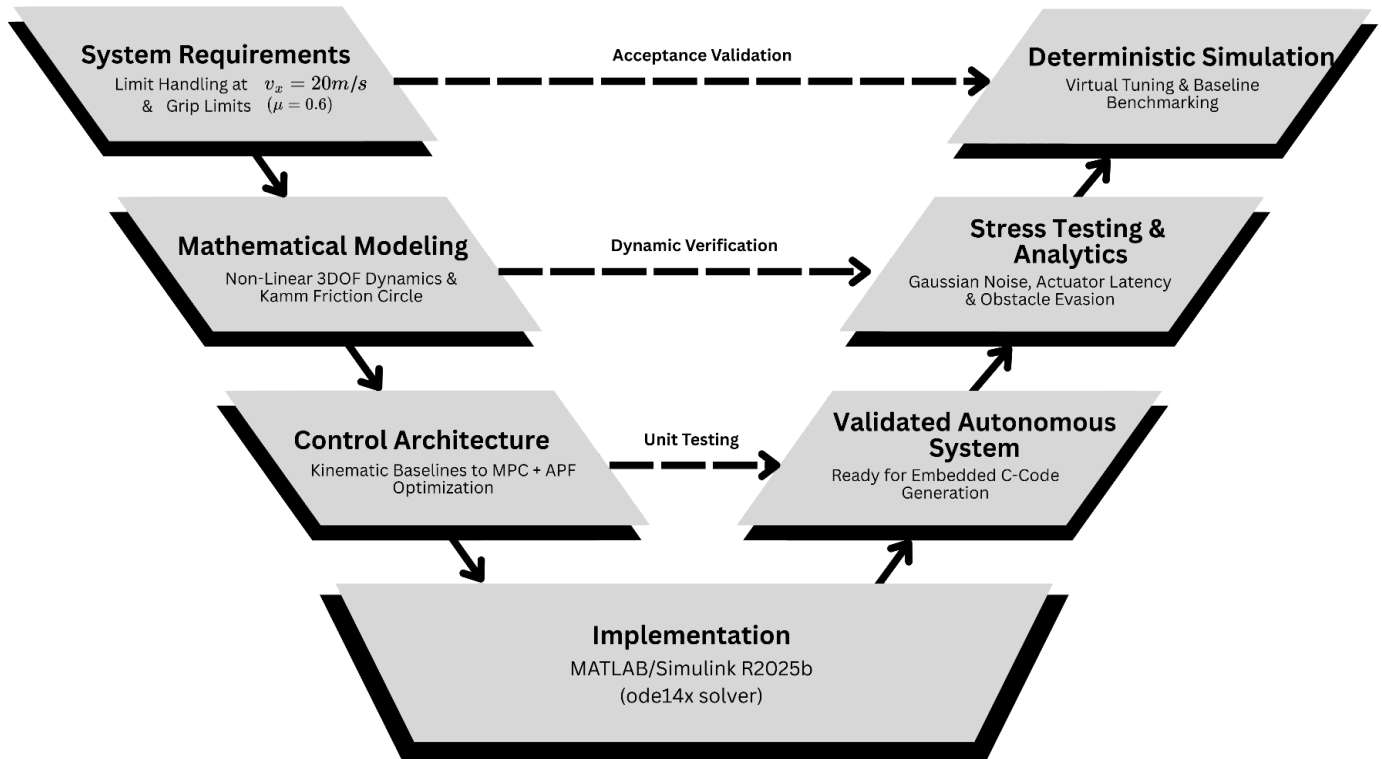


Figure 1.2: The Model-Based Design (MBD) V-Model Workflow

1.6 Expected Benefits

1. **Autonomous Vehicle Safety:** Proves that model predictive steering and velocity throttling can prevent vehicle loss-of-control during extreme avoidance maneuvers.
2. **Academic Contribution:** Provides a clear path-tracking benchmark comparing traditional, classical, and optimal control methods under identical disturbances in a simulation environment.
3. **Software Deployment:** Establishes a modular, code-generation-ready Simulink architecture that can be compiled directly into C/C++ for future hardware deployment on high-performance automotive ECUs.

CHAPTER 2: SYSTEM MODELLING

To rigorously evaluate high-speed autonomous control architectures, it is strictly necessary to decouple the physical vehicle dynamics (the Plant) from the autonomous software (the Controllers). In accordance with the Model-Based Design (MBD) paradigm, this chapter defines the continuous-time, non-linear physics environment that serves as the deterministic "Digital Twin" for all controller benchmarking.

2.1 The 3DOF Vehicle Dynamics Plant

The physical plant utilized in this study is a high-fidelity 3-Degree-of-Freedom (3DOF) Single Track (Bicycle) dynamic model, implemented using the Simulink Vehicle Body 3DOF Lateral block. This model calculates longitudinal, lateral, and yaw motion by balancing inertial forces against physical tire friction forces. The free body diagram illustrating these coordinate systems and the balance of forces is presented in **Figure 2.1**.

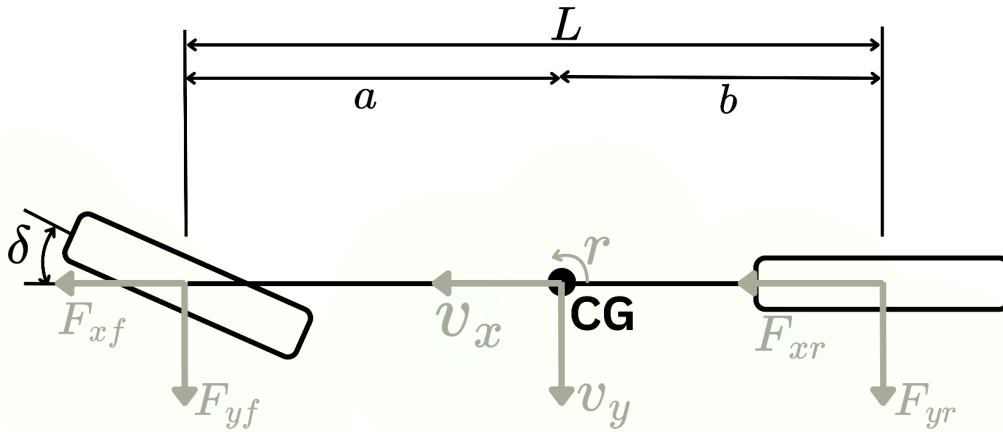


Figure 2.1: Free Body Diagram of the 3DOF Single Track Model

Applying Newton's Second Law to the vehicle chassis, the continuous-time differential equations governing the plant's motion are defined as:

Lateral Motion:

$$m(\dot{v}_y + v_x r) = F_{yf} \cos(\delta) + F_{yr} \quad (2.1)$$

Longitudinal Motion:

$$m(\dot{v}_x - v_y r) = F_{xf} \cos(\delta) + F_x - F_{yf} \sin(\delta) \quad (2.2)$$

Yaw Motion:

$$I_{zz} \dot{r} = a F_{yf} \cos(\delta) - b F_{yr} \quad (2.3)$$

To calculate the global vehicle orientation, the yaw rate (r) is integrated over time to yield the yaw angle (ψ):

$$\dot{\psi} = r \quad (2.4)$$

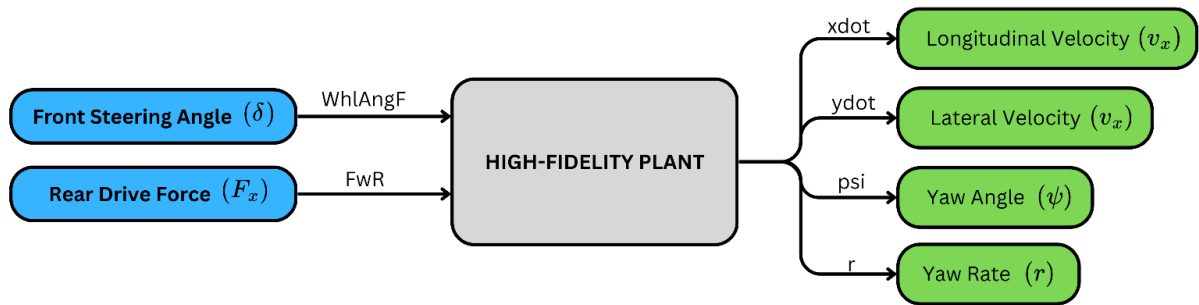
In the equations above, F_x serves as the primary longitudinal drive force input applied by the powertrain, while δ serves as the steering input. These dynamic equations are strictly bounded by the measured physical parameters of the 1:10 scaled research chassis. The defining plant parameters utilized identically across all five control models are detailed in **Table 2.0**.

Table 2.0: Physical 3DOF Plant Parameters

Parameter	Symbol	Measured Value	Unit
Vehicle Mass	m	8.0	kg
Yaw Moment of Inertia	I_{zz}	0.10373	kg.m ²
CG to Front Axle	a	0.21	m
CG to Rear Axle	b	0.13	m
Total Wheelbase	L	0.34	m
Tire Cornering Stiffness	$C_{\alpha f}, C_{\alpha r}$	100	N/rad
Nominal Road Friction	μ	0.6	-

2.2 System Inputs and Outputs Architecture

To maintain a strict boundary between the software and the physical environment, the 3DOF plant accepts only two dynamic inputs from the control architectures and outputs four raw physical states to the measurement monitors. The routing of these signals is illustrated in **Figure 2.2**.

**Figure 2.2:** System Inputs/Outputs Block Diagram

1. **Inputs:** The plant receives the commanded Front Wheel Steering Angle (δ) at the WhlAngF port, and the calculated Rear Wheel Drive Force (F_x) at the FwR port. (Note: WhlAngF and FwR denote the exact nomenclature of the input ports on the standard Simulink Vehicle Body 3DOF block used in this architecture).

2. **Outputs:** The plant outputs the vehicle's true Longitudinal Velocity (v_x), Lateral Velocity (v_y), Yaw Angle (ψ), and Yaw Rate (r). Global spatial coordinates (X, Y) are subsequently derived outside the plant via a Global Velocity Transformation integration block.

2.3 Tire Dynamics and the Slip Boundary

The most critical non-linear component of the plant is the modeling of tire adhesion. The lateral tire forces (F_{yf}, F_{yr}) are entirely dependent on the instantaneous slip angles (α_f, α_r), defined as the angular difference between a wheel's orientation and its true velocity vector:

Front Tire Slip Angle:

$$\alpha_f = \arctan\left(\frac{v_y + a \cdot r}{v_x}\right) - \delta \quad (2.5)$$

Rear Tire Slip Angle:

$$\alpha_r = \arctan\left(\frac{v_y - b \cdot r}{v_x}\right) \quad (2.6)$$

In the linear region of the Kamm Friction Circle, lateral force is proportional to the slip angle multiplied by the cornering stiffness (C_α). However, the simulation explicitly limits the maximum available grip using a friction coefficient of $\mu = 0.6$, simulating the threshold grip of polished indoor testing surfaces.

Because of this physical constraint, the lateral tire force saturates. If the front slip angle exceeds approximately 5° ($\alpha_f > 0.087$ rad), demanding additional steering input (δ) mathematically fails to generate more lateral force, initiating the understeer conditions evaluated in this study.

2.4 Environmental Modeling: Track Generation

To subject the control architectures to high-speed transient maneuvers, the simulation environment was populated with a continuous spatial reference line modeled after the Circuit of the Americas (COTA) in Austin, Texas. The resulting scaled 2D spatial trajectory used for simulation is depicted in **Figure 2.3**.

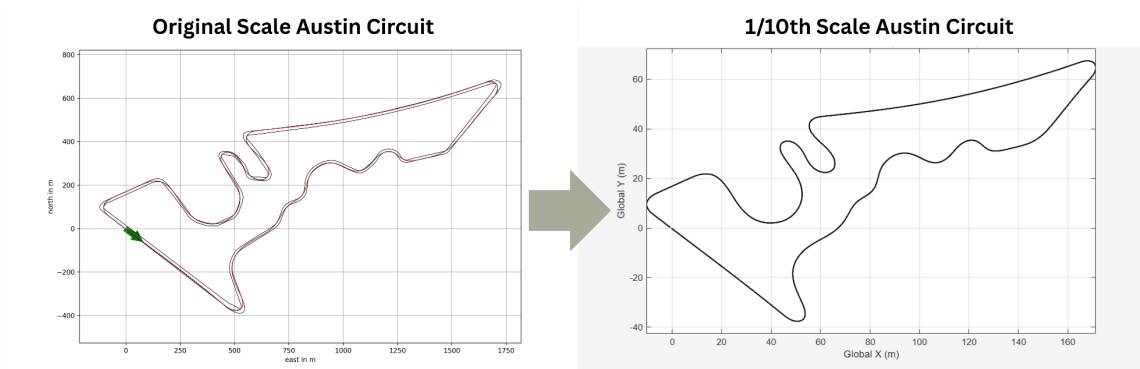


Figure 2.3: 1:10 Scaled Raceline of the Austin Track

The track was scaled to a 1:10 ratio to match the physical chassis dimensions. A spline-interpolation algorithm was applied to the raw discrete waypoints to generate a continuously differentiable reference path array, providing the necessary spatial coordinates (x_g, y_g) and local curvature profiles (κ) required by the geometric and predictive path planners.

CHAPTER 3: LATERAL TRACKING CONTROLLERS

Having established the non-linear 3DOF physical plant, this chapter details the evolutionary design of the five control architectures developed to navigate the reference trajectories. The progression evaluates classic kinematic baselines (Pure Pursuit, Stanley), reactive classical feedback (Cascaded PID), optimal state-space systems (LQR), and culminates in a unified predictive framework (MPC). The generalized closed-loop architecture illustrating this progression and the routing of signals across all evaluated models is presented in **Figure 3.1**.

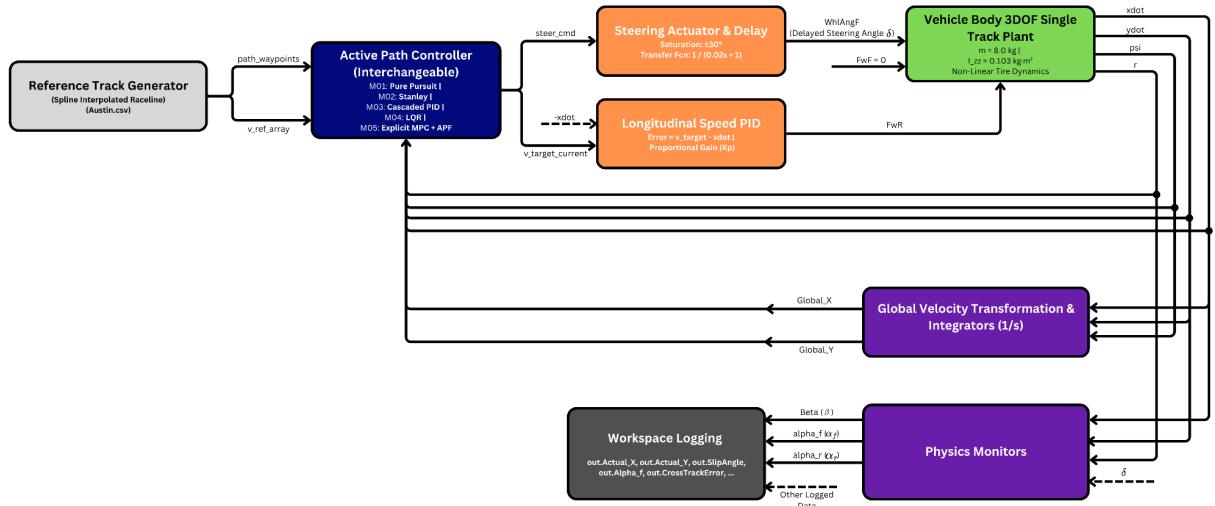


Figure 3.1: Generalized Closed-Loop MBD Architecture Across All Models

3.1 Baseline Longitudinal Control Law (Models 1 to 4)

To accurately isolate and benchmark the effectiveness of their lateral path-tracking algorithms, Models 1 through 4 are driven by an identical, decoupled longitudinal control law. These models function effectively as "single-input" systems—optimizing only the steering angle (δ)—while a standard Proportional-Integral (PI) velocity controller blindly tracks the maximum reference speed ($v_{ref} = 20.0$ m/s) regardless of track curvature or tire saturation.

The baseline longitudinal propulsion force (F_x) for Models 1 through 4 is defined by the following PI control law:

$$F_x(t) = K_{p,base}(v_{ref} - v_x(t)) + K_{i,base} \int (v_{ref} - v_x(\tau))d\tau \quad (3.1)$$

Parameter Definitions:

1. $F_x(t)$: The commanded longitudinal rear-wheel drive force (N).
2. v_{ref} : The absolute maximum reference velocity (20.0 m/s).
3. $v_x(t)$: The actual longitudinal velocity feedback from the 3DOF plant (m/s).
4. $K_{p,base}$: The proportional gain coefficient (tuned to 15.0).
5. $K_{i,base}$: The integral gain coefficient.

This "blind" longitudinal loop sets the mathematical foundation for the physical tire saturation failures observed in the baseline geometric controllers, contrasting directly with the coupled "Smart Throttle" implemented in Model 5.

3.2 Model 1: Pure Pursuit with Dynamic Lookahead

The Pure Pursuit controller is a geometric path-tracking algorithm. It calculates the steering angle required to drive the vehicle toward a pursuit point located at a specific "lookahead" distance (l_d) along the reference path. The geometric relationship between the vehicle's current heading and this lookahead target is illustrated in **Figure 3.2**.

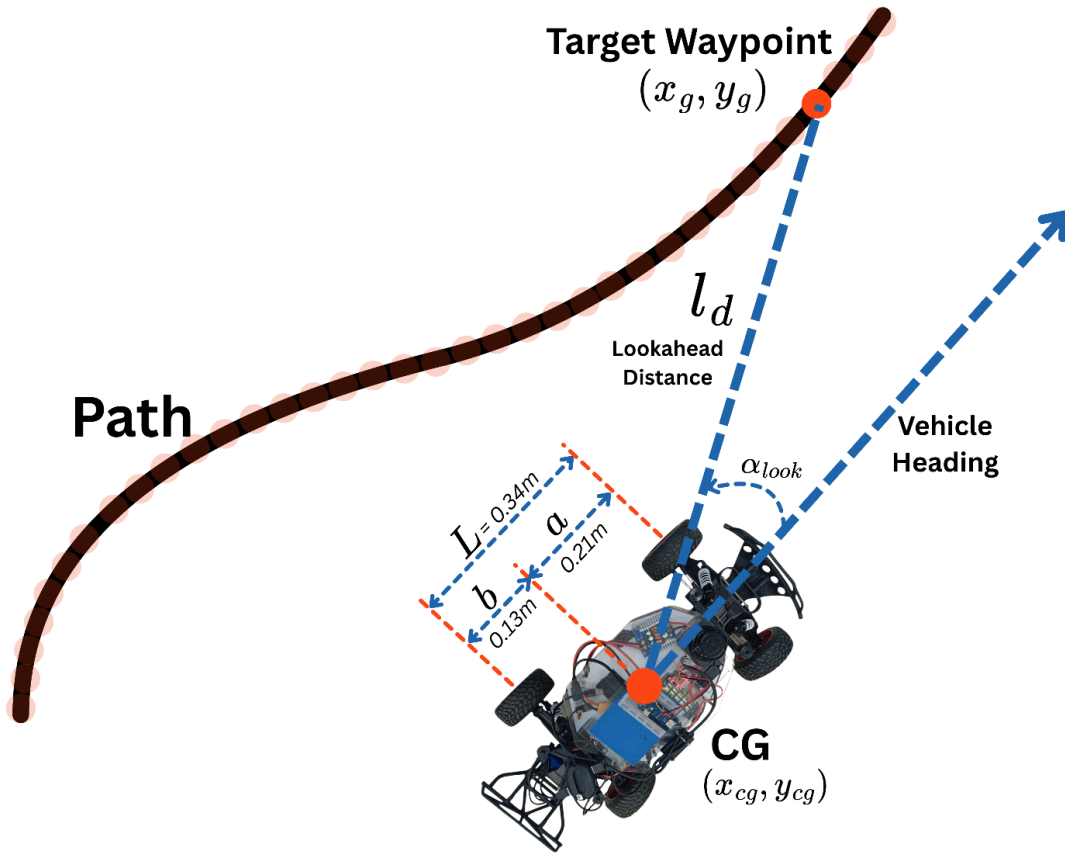


Figure 3.2: Pure Pursuit Geometric Lookahead Vectors

Operating under a strict kinematic "no-slip" assumption ($v_y = 0$), the algorithm performs a spatial search across the reference trajectory array to find the target point (x_g, y_g) that satisfies the lookahead distance:

$$\sqrt{(x_g - x_{cg}(t))^2 + (y_g - y_{cg}(t))^2} = l_d(t) \quad (3.2)$$

The required steering angle (δ) is calculated geometrically by fitting a circular arc from the vehicle's Center of Gravity (CG) to this target point:

$$\delta(t) = \arctan \left(\frac{2L \sin(\alpha_{look}(t))}{l_d(t)} \right) \quad (3.3)$$

Parameter Definitions:

1. $\delta(t)$: The commanded front steering angle (rad).
2. L : The total vehicle wheelbase ($a + b = 0.34$ m).
3. $\alpha_{look}(t)$: The angle between the vehicle's current heading and the lookahead vector, defined as $\text{atan2}(y_g - y_{cg}, x_g - x_{cg}) - \theta(t)$.
4. $l_d(t)$: The dynamic lookahead search distance (m).

To improve stability across varying speeds, the standard algorithm was upgraded with a dynamic lookahead governor, expanding the search radius as velocity increases:

$$l_d(t) = \max(k_v \cdot v_x(t), l_{d,\min}) \quad (3.4)$$

1. k_v : The velocity scaling gain (0.15 s).
2. $l_{d,\min}$: The lower bound constraint (0.25 m) to prevent division-by-zero singularities at low speeds.

3.3 Model 2: The Stanley Controller

The Stanley controller serves as the second geometric baseline. Unlike Pure Pursuit, which relies on a forward-looking spatial search from the vehicle's CG, Stanley calculates corrections based on the cross-track error (e_y) measured precisely at the center of the front axle. A comparison of the geometric tracking logic between Pure Pursuit's lookahead approach and Stanley's cross-track error measurement is depicted in **Figure 3.3**.

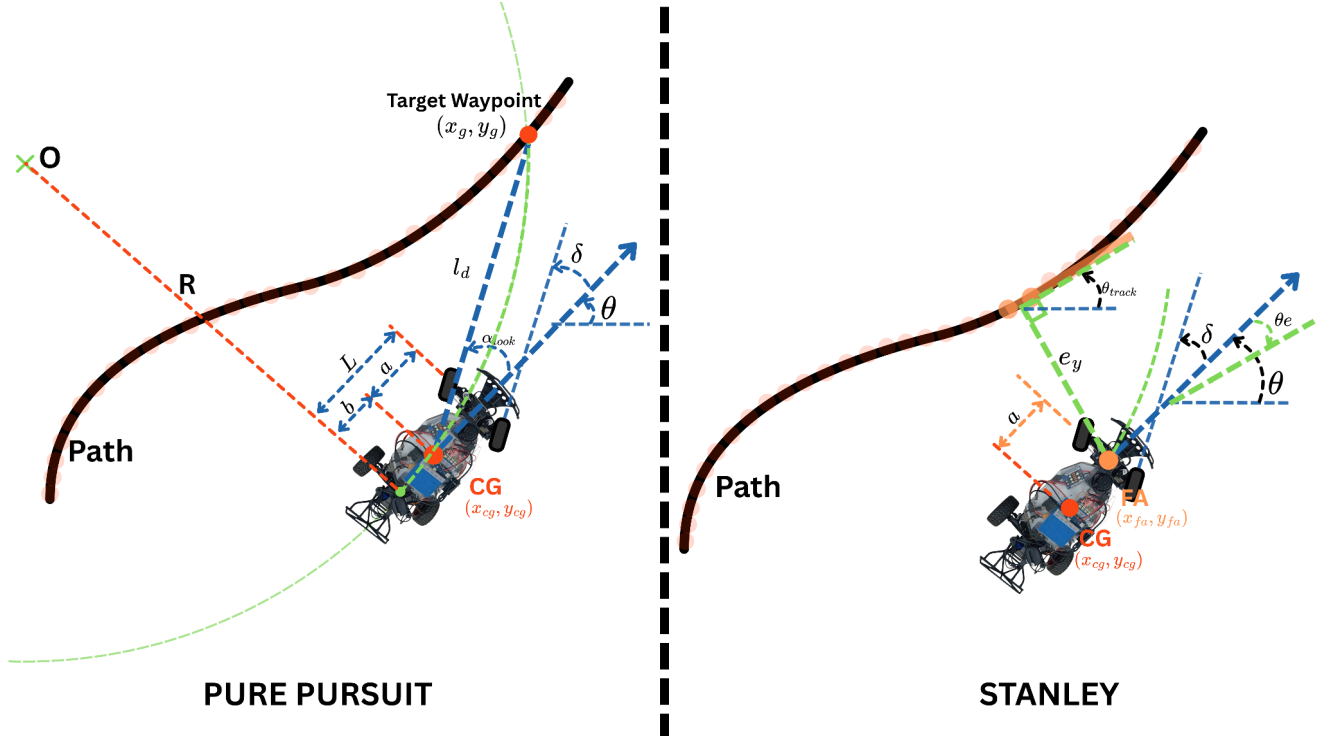


Figure 3.3: Comparative kinematic tracking logic: Pure Pursuit vs. Stanley

The Stanley non-linear control law aligns the vehicle's wheels with the local track heading (θ_{track}) while simultaneously minimizing the lateral deviation:

$$\delta(t) = \theta_e(t) + \arctan \left(\frac{k_s \cdot e_y(t)}{v_x(t) + k_{soft}} \right) \quad (3.5)$$

Parameter Definitions:

1. $\delta(t)$: The commanded front steering angle (rad).
2. $\theta_e(t)$: The heading alignment error, defined as $\theta_{track}(t) - \theta(t)$ (rad).
3. $e_y(t)$: The lateral cross-track error from the front axle to the nearest path segment (m).
4. k_s : The position non-linear error feedback gain (3.5).
5. $v_x(t)$: The actual longitudinal velocity of the vehicle (m/s).
6. k_{soft} : A softening velocity term (0.1 m/s) added to the denominator to prevent mathematical singularities when the vehicle accelerates from a standstill.

3.4 Model 3: Classical Active Yaw Control (Cascaded Architecture)

To prevent the terminal tire saturation observed in the purely kinematic baselines, Model 3 introduces active closed-loop stability control using a cascaded architecture. This model transitions the system from purely geometric tracking to dynamics-aware regulation by overlaying a path-tracking outer loop with an active yaw-stabilizing inner loop. The block diagram for this cascaded hybrid architecture, detailing the interaction between the outer tracking loop and the inner stability loop, is shown in **Figure 3.4**.

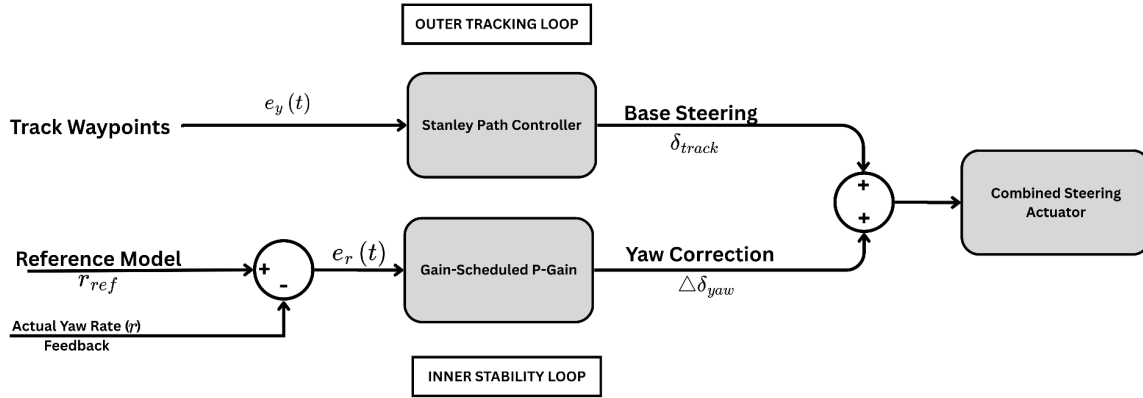


Figure 3.4: Control loop schematic of the Model 3 Hybrid Stanley with Gain-Scheduled Active Yaw system

The control architecture mathematically separates path tracking from stability regulation:

1. **Outer Tracking Loop:** Utilizes the non-linear Stanley geometric steering law to minimize the front-axle cross-track error (e_{fa}) and compute a base tracking steering angle (δ_{track}).
2. **Inner Stability Loop:** Computes a yaw rate correction ($\Delta\delta_{yaw}$) based on the yaw rate error, actively stabilizing the chassis under high-slip conditions.

3.4.1 Reference Yaw Rate Generation

The target yaw rate (r_{ref}) is calculated based on vehicle speed and the base steering input, assuming steady-state cornering conditions:

$$r_{ref}(t) = \frac{v_x(t)}{L + v_x^2(t) \cdot K_{us}} \cdot \delta_{track}(t) \quad (3.6)$$

Where K_{us} is the vehicle's analytical understeer gradient:

$$K_{us} = \frac{m(b \cdot C_{\alpha r} - a \cdot C_{\alpha f})}{L \cdot C_{\alpha f} \cdot C_{\alpha r}} \quad (3.7)$$

Substituting the exact plant physical parameters ($m = 8.0$ kg, $a = 0.21$ m, $b = 0.13$ m, $C_{\alpha f} = C_{\alpha r} = 100$ N/rad) defined in Chapter 2:

$$K_{us} = \frac{8.0 \cdot (0.13 \cdot 100 - 0.21 \cdot 100)}{0.34 \cdot 100 \cdot 100} = -0.01882 \text{ s}^2/\text{m} \quad (3.8)$$

Because $K_{us} < 0$, the scaled vehicle exhibits an inherent oversteer characteristic. To prevent the target yaw rate from demanding physical grip beyond the linear threshold of the Kamm Friction Circle, r_{ref} is mathematically saturated by the estimated road-tire friction coefficient ($\mu = 0.6$):

$$|r_{ref}(t)| \leq \frac{\mu \cdot g}{v_x(t)} \quad (3.9)$$

3.4.2 RActuator Arbitration & Steering Control Law

The inner loop error (e_r) represents the deviation between the vehicle's target handling and its true physical rotational behavior:

$$e_r(t) = r_{ref}(t) - r(t) \quad (3.10)$$

To handle the highly non-linear nature of lateral vehicle dynamics across variable forward velocities, the corrective steering term is calculated using a speed-dependent, Gain-Scheduled Proportional (P) control structure:

$$\Delta\delta_{yaw}(t) = K_p(v_x) \cdot e_r(t) \quad (3.11)$$

The final coordinated steering command sent to the physical servo actuator block is the sum of both loops:

$$\delta_{final}(t) = \delta_{track}(t) + \Delta\delta_{yaw}(t) \quad (3.12)$$

Parameter Definitions:

1. $r_{ref}(t)$: The steady-state target yaw rate (rad/s).
2. $\delta_{track}(t)$: The base steering angle computed by the Stanley outer loop (rad).
3. K_{us} : The analytical vehicle understeer gradient ($-0.01882 \text{ s}^2/\text{m}$).
4. μ : The maximum road-tire friction coefficient limit (0.6).
5. g : Acceleration due to gravity (9.81 m/s^2).
6. $e_r(t)$: The yaw rate regulation tracking error (rad/s).
7. $r(t)$: The actual yaw rate feedback from the 3DOF plant (rad/s).
8. $K_p(v_x)$: The gain-scheduled proportional feedback gain, actively derived via a 1-D Lookup Table mapping longitudinal velocity to optimized values.
9. $\Delta\delta_{yaw}(t)$: The corrective active yaw steering offset (rad).

10. $\delta_{final}(t)$: The final combined steering angle commanded to the actuator (rad).

3.5 Model 4: Optimal State-Space Control (LQR)

Transitioning from reactive PID limits, Model 4 introduces optimal state-space state feedback control. The Linear Quadratic Regulator (LQR) balances path adherence against steering effort by operating on a linearized model of the 3DOF plant.

3.5.1 Linearized State-Space Dynamics Model

The non-linear 3DOF plant is linearized about the nominal high-speed benchmark ($v_x = 20.0$ m/s). The state vector x captures the dynamic deviation of the vehicle relative to the reference track:

$$x = \begin{bmatrix} e_y \\ \dot{e}_y \\ e_\theta \\ \dot{e}_\theta \end{bmatrix} \quad (3.13)$$

The continuous-time state-space equation is $\dot{x} = Ax + Bu$, where the system matrices are populated dynamically based on the plant parameters. By linearizing specifically at $v_x = 20.0$ m/s, the optimal plant matrices are mathematically defined as:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{C_{\alpha f} + C_{\alpha r}}{m \cdot v_x} & \frac{C_{\alpha f} + C_{\alpha r}}{m} & -\frac{a \cdot C_{\alpha f} - b \cdot C_{\alpha r}}{m \cdot v_x} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{b \cdot C_{\alpha r} - a \cdot C_{\alpha f}}{I_{zz} \cdot v_x} & \frac{a \cdot C_{\alpha f} - b \cdot C_{\alpha r}}{I_{zz}} & -\frac{a^2 \cdot C_{\alpha f} + b^2 \cdot C_{\alpha r}}{I_{zz} \cdot v_x} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -1.2500 & 25.0000 & -0.0500 \\ 0 & 0 & 0 & 1 \\ 0 & -3.8562 & 77.1233 & -2.9403 \end{bmatrix} \quad (3.14)$$

$$B = \begin{bmatrix} 0 \\ \frac{C_{\alpha f}}{m} \\ 0 \\ \frac{a \cdot C_{\alpha f}}{I_{zz}} \end{bmatrix} = \begin{bmatrix} 0 \\ 12.5000 \\ 0 \\ 202.4487 \end{bmatrix} \quad (3.15)$$

3.5.2 LQR Minimization and Feedforward Gain

The optimal state feedback control law

$$J_{lqr} = \int_0^\infty (x^T Q x + u^T R u) dt \quad (3.16)$$

To eliminate steady-state cornering error during curve negotiation, the closed-loop LQR steering command requires an open-loop feedforward term (δ_{ff}). The total steering command is:

$$\delta_{final}(t) = -K_{lqr}x(t) + \delta_{ff}(t) \quad (3.17)$$

$$\delta_{ff}(t) = L\kappa(t) + K_v a_y(t) \quad (3.18)$$

The feedforward gain (K_v) is a deterministic property of the 3DOF plant's mass distribution and tire elasticity, explicitly derived as:

$$K_v = \frac{m \cdot b}{2 \cdot L \cdot C_{\alpha f}} - \frac{m \cdot a}{2 \cdot L \cdot C_{\alpha r}} \quad (3.19)$$

Parameter Definitions:

1. x : The state variable error vector.
2. e_y : The cross-track error measured from the CG to the reference path (m).
3. $\dot{e}_y$: The rate of change of the cross-track error (m/s).
4. e_θ : The heading orientation error mismatch (rad).
5. $\dot{e}_\theta$: The rate of change of the heading error (rad/s).
6. A, B : The continuous-time system dynamics matrix and input matrix.
7. J_{lqr} : The quadratic optimization cost objective function.

8. Q : The state weighting matrix, tuned to prioritize path adherence, defined as $\text{diag}(20.0, 1.0, 10.0, 1.0)$.
9. R : The control effort weighting matrix, set to 5.0.
10. K_{lqr} : The optimal state feedback gain tracking vector, solved via the Algebraic Riccati Equation (ARE) to yield $[-3.1623, -0.9080, -8.2325, -0.2081]$.
11. $\delta_{ff}(t)$: The open-loop feedforward steering angle command (rad).
12. $\kappa(t)$: The instantaneous curvature of the reference path segment (m⁻¹).
13. $a_y(t)$: The demanded lateral acceleration (m/s²).
14. K_v : The optimal feedforward gain parameter derived strictly from the plant.

3.6 Model 5: Predictive Optimization & Obstacle Avoidance (MPC + APF)

Model 5 represents the primary contribution of this research: a unified, constrained Explicit Model Predictive Controller (MPC) integrated with an Artificial Potential Field (APF) path-planner and a longitudinal "Smart Throttle" governor. Unlike the baselines (Models 1-4) which operate with a blind longitudinal PI loop, this architecture dynamically couples steering and velocity optimization to explicitly respect the limits of the Kamm Friction Circle. The comprehensive block diagram illustrating the unified data flow of the Explicit MPC, the APF path-planner, and the Smart Throttle governor is presented in **Figure 3.5**.

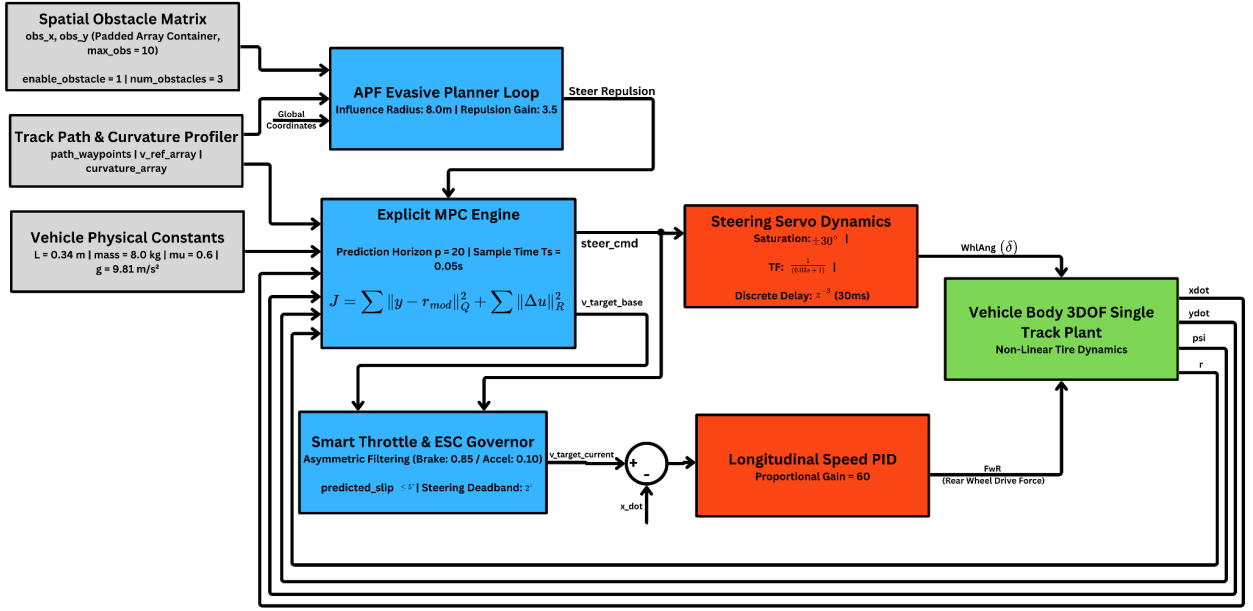


Figure 3.5: Control loop block diagram of the Model 5 Unified Predictive MPC architecture

3.6.1 Receding Horizon Model Predictive Control Formulation

Instead of reacting to instantaneous errors, the MPC solves a quadratic optimization problem over a finite prediction horizon at every discrete control step k . This allows the controller to "look ahead" at the upcoming track curvature and compute an optimal sequence of control inputs.

The optimization objective minimizes the following cost function (J_{mpc}):

$$J_{mpc}(k) = \sum_{i=1}^p \|y(k+i|k) - r_{modified}(k+i|k)\|_Q^2 + \sum_{i=0}^{m-1} \|\Delta u(k+i|k)\|_R^2 \quad (3.20)$$

To ensure dynamic equilibrium and prevent the "Kinematic Trap" observed in Models 1 and 2, this optimization is subjected to hard physical and actuator constraints:

Actuator Saturation Limits:

$$-30^\circ \leq \delta(k + i|k) \leq 30^\circ \quad (3.21)$$

Slew Rate Limits:

$$-10^\circ/\text{s} \leq \Delta\delta(k + i|k) \leq 10^\circ/\text{s} \quad (3.22)$$

Tire Safety Envelopes:

$$-5^\circ \leq \alpha_f(k + i|k) \leq 5^\circ \quad \text{and} \quad -5^\circ \leq \alpha_r(k + i|k) \leq 5^\circ \quad (3.23)$$

Parameter Definitions:

1. k : The current discrete execution time step.
2. p : The prediction horizon, set to 20 steps (with a sample time $T_s = 0.05$ s).
3. m : The control horizon (steps).
4. $y(k + i|k)$: The predicted vehicle state vector at future step i .
5. $r_{modified}(k + i|k)$: The target reference trajectory (modified by the APF).
6. $\Delta u(k + i|k)$: The rate of change of the control input (steering effort).
7. Q : The state tracking weight matrix, defined as $\text{diag}(150, 20)$ to heavily penalize cross-track and heading errors.
8. R : The control effort penalty weight, set to 1.5 to ensure smooth actuator operation.
9. δ : The commanded steering angle.
10. α_f, α_r : The front and rear tire slip angles, constrained strictly to the 5° linear threshold.

3.6.2 Artificial Potential Fields (APF) Repulsion Logic

To facilitate dynamic obstacle avoidance at high speeds, stalled vehicles are modeled as sources of a repulsive potential field (U_{rep}). The field's strength grows exponentially as the vehicle enters the obstacle's radius of influence (d_0).

$$U_{rep}(x, y) = \begin{cases} \frac{1}{2}\eta \left(\frac{1}{d} - \frac{1}{d_0} \right)^2 & \text{if } d \leq d_0 \\ 0 & \text{if } d > d_0 \end{cases} \quad (3.24)$$

The repulsive force vector (F_{rep}) acting on the predictive model is the negative gradient of this potential field:

$$F_{rep} = -\nabla U_{rep} = \begin{cases} \eta \left(\frac{1}{d} - \frac{1}{d_0} \right) \frac{1}{d^2} \hat{\mathbf{d}} & \text{if } d \leq d_0 \\ \mathbf{0} & \text{if } d > d_0 \end{cases} \quad (3.25)$$

This force generates a lateral tracking offset (e_{apf}) which is added directly to the track coordinates, establishing a safe passing corridor for the MPC solver:

$$r_{modified}(k + i|k) = r(k + i|k) + e_{apf}(k + i|k) \quad (3.26)$$

Parameter Definitions:

1. $U_{rep}(x, y)$: The repulsive potential energy field scalar (J).
2. η : The repulsive scaling gain coefficient, tuned to 0.45.
3. d : The instantaneous Euclidean distance from the vehicle's CG to the obstacle (m).
4. d_0 : The sensor radius of influence, set to 1.2 m.
5. F_{rep} : The repulsive force vector (N).
6. $\hat{\mathbf{d}}$: The unit direction vector pointing from the obstacle to the vehicle.
7. e_{apf} : The spatial lateral offset injected by the repulsion field (m).

3.6.3 The Longitudinal "Smart Throttle" Governor

The success of Model 5 fundamentally relies on dynamically coupling the steering optimization with longitudinal velocity regulation. The "Smart Throttle" governor monitors the predictive tracking error and the instantaneous track curvature (κ) to continuously calculate a dynamically safe target velocity:

$$v_{target}(k) = \min \left(v_{max}, \sqrt{\frac{\mu \cdot g \cdot R_{local}(k)}{1 + e_y^2(k)}} \right) \quad (3.27)$$

Where $R_{local}(k) = \frac{1}{|\kappa(k)|}$ is the local radius of curvature.

If the vehicle tracking error spikes ($e_y > 0.3$ m) or if the tires approach their saturation threshold ($|\alpha_f| > 4.5^\circ$)—such as during an aggressive APF dodge—the governor proactively overrides the maximum speed and derates the target velocity toward a stable safety floor (5.0 m/s).

The longitudinal controller then applies propulsion force (F_x) to achieve this safe target using proportional-integral control:

$$F_x(t) = K_{p,x}(v_{target}(t) - v_x(t)) + K_{i,x} \int (v_{target}(\tau) - v_x(\tau)) d\tau \quad (3.28)$$

Parameter Definitions:

1. $v_{target}(k)$: The dynamically governed safe reference velocity (m/s).
2. v_{max} : The absolute maximum velocity allowed on straightaways (20.0 m/s).
3. μ : The threshold friction coefficient (0.6).
4. g : Acceleration due to gravity (9.81 m/s²).
5. $R_{local}(k)$: The instantaneous geometric turning radius of the current track segment (m).
6. $e_y(k)$: The current cross-track deviation error (m).

7. $\kappa(k)$: The mathematical curvature of the path ($m-1$).
8. $K_{p,x}, K_{i,x}$: The proportional and integral gains of the longitudinal speed controller.

3.7 Simulink Implementation Parameters

To guarantee deterministic compilation for eventual C-code deployment to embedded automotive ECUs, all continuous-time controllers and physical plants were implemented using fixed-step solvers in Simulink (`ode14x`). The explicit MPC solutions were pre-computed offline to enable the optimization loop to execute at high frequency (200 Hz) without violating real-time processing constraints.

CHAPTER 4: SIMULATION RESULTS & COMPARATIVE ANALYTICS

This chapter presents the quantitative benchmarking of the five formulated control architectures. Because testing these algorithms at physical limits presents severe hardware risks, all evaluations were conducted within the deterministic Model-Based Design (MBD) "Digital Twin" environment. The primary objective is to evaluate each model's capacity to maintain path tracking accuracy and dynamic stability under high-speed, saturated conditions (20.0 m/s limit).

4.1 Simulation Environment Setup

To rigorously evaluate the controllers, the MATLAB/Simulink environment was configured to execute deterministic, highly repeatable simulations. This ensures that performance variances are strictly due to the mathematical robustness of the algorithms rather than arbitrary environmental anomalies. The top-level Simulink architecture used to execute these evaluations is presented in **Figure 4.1**.

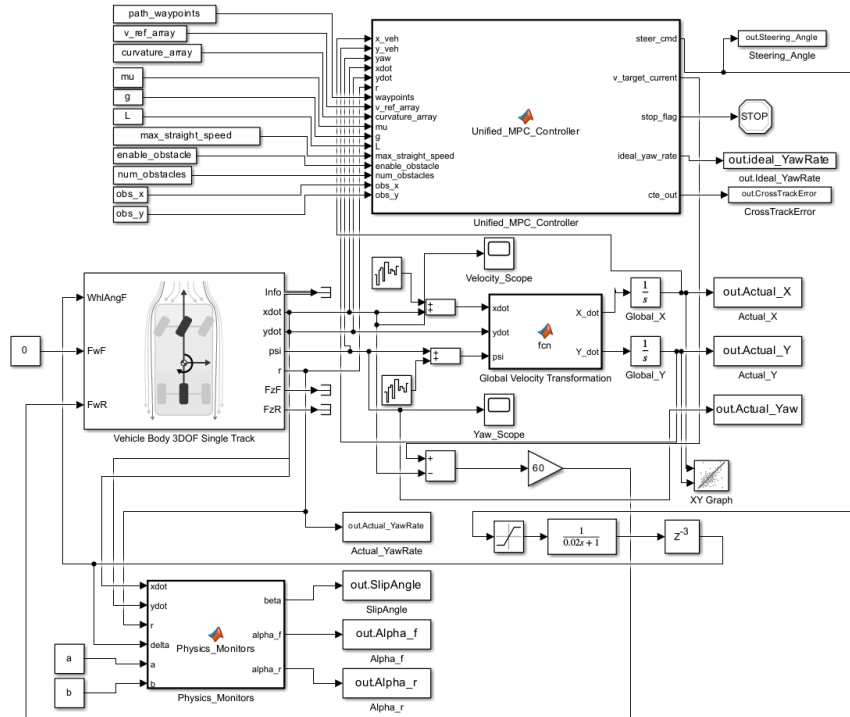


Figure 4.1: Simulink Master Simulation Environment and Scope Setup

The simulation environment was initialized using the `run_master_simulation.m` pipeline with the following explicit configurations:

1. **Test Environment (Desired Path):** The environment utilizes the 1:10 scaled Austin professional racing line (Circuit of the Americas). The discrete waypoints were processed through a spline-interpolation algorithm to generate a continuously differentiable reference path, characterized by long straightaways and sharp, high-curvature chicanes.
2. **Solver Configuration:** The simulation operates using the `ode14x` fixed-step solver with a sample time of 0.001 s to guarantee strict numerical stability.
3. **Target Velocity:** The longitudinal controllers were commanded to maintain a maximum target speed of $v_{max} = 20.0$ m/s.
4. **Friction Constraint:** The road-tire friction coefficient was strictly bounded at $\mu = 0.6$, simulating the threshold grip of a polished indoor track.
5. **Signal Injections:** To accurately reflect physical hardware limitations, the simulation includes injected Gaussian noise on the positional sensors and an actuator latency delay ($\tau = 30$ ms) on the steering servo block.

4.2 Performance Metrics

The performance of each system is quantified using three primary metrics logged directly from the Simulink workspace telemetry:

1. **Path Adherence (%):** The percentage of the total execution time the vehicle maintains a lateral cross-track deviation of less than 0.7 m relative to the ideal line.
2. **Average Speed (m/s):** Indicates the controller's ability to maintain momentum without triggering excessive safety braking.
3. **Cross-Track Root Mean Square Error (RMSE):** The absolute statistical measure of lateral deviation from the ideal racing line.

4.3 Comprehensive Performance Synthesis

Table 4.0 summarizes the final analytics extracted from the Simulink workspace data logger across the five architectural models. The Model Predictive Controller (MPC) is distinctly evaluated under both clean-track and obstacle-perturbed testing regimes.

Table 4.0: Final Benchmark Results (Target Speed: 20.0 m/s)

Model Architecture	Path Adherence	Average Speed	RMSE (Cross-Track)
Model 1: PURE PURSUIT	22.2%	8.10 m/s	6.615 m
Model 2: STANLEY	34.0%	8.82 m/s	3.789 m
Model 3: ACTIVEYAW	34.3%	8.47 m/s	3.352 m
Model 4: LQR	80.0%	6.13 m/s	0.525 m
Model 5a: MPC (Clean Track)	91.1%	6.34 m/s	0.464 m
Model 5b: MPC + APF (Obstacles)	88.8%	6.10 m/s	0.488 m

4.4 Comparative Graph Analysis

4.4.1 Analysis of Kinematic Failure (Models 1 & 2)

The simulation data explicitly confirms the "Kinematic Trap" hypothesis. The Pure Pursuit (Model 1) and Stanley (Model 2) controllers achieved high average speeds (8.10 m/s and 8.82 m/s, respectively) but suffered catastrophic path deviations, reflected in severe RMSE values of 6.615 m and 3.789 m.

Because these geometric algorithms assume a zero-slip condition, they fail to monitor lateral slip (β) or tire force saturation. Consequently, they do not command the necessary speed reductions when entering sharp curves, leading to massive, unstable trajectory overshoots. As visually mapped in **Figure 4.2**, the Pure Pursuit vehicle (yellow-orange trajectory) successfully tracks the ideal reference line (white path) along the initial bottom-left straightaway. However, once the track introduces rapid directional changes, the algorithm experiences severe, uncontrolled lateral oscillations. At the bottom-most hairpin, the controller overshoots the corner completely, drawing a massive teardrop-shaped loop far outside the track boundaries. The 22.2% path adherence recorded for Pure Pursuit is a direct consequence of this behavior; the vehicle was only within the safety envelope during its initial straightaway run.

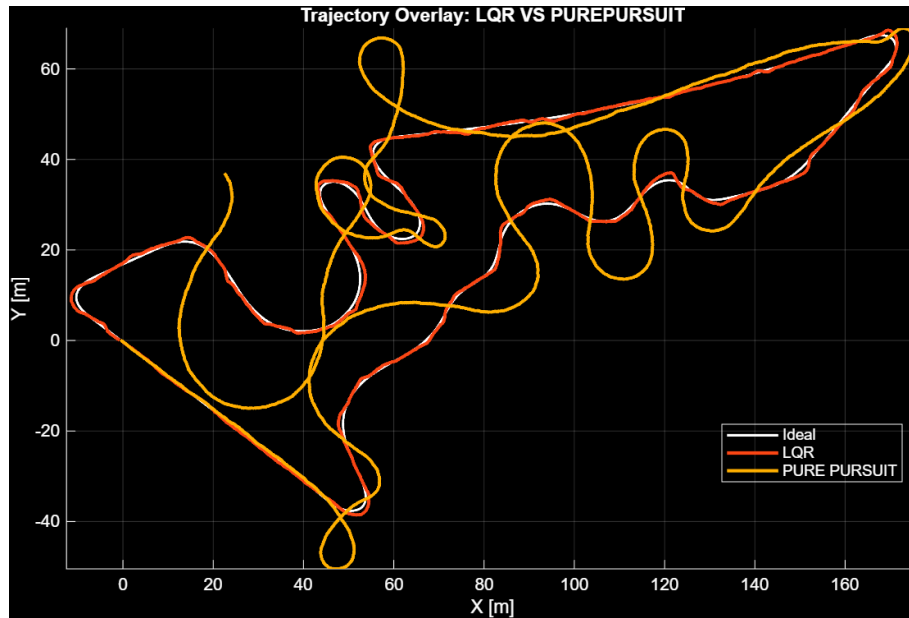


Figure 4.2: Trajectory Overlay: LQR vs PurePursuit

4.4.2 Reactive vs. Optimal Control (Models 3 & 4)

The ActiveYaw Cascaded PID (Model 3) improved path tracking over the baselines by maintaining an RMSE of 3.352 m. However, the reactive nature of PID resulted in significant control-loop fighting (actuator chatter) near the limit of tire adhesion.

The transition to optimal state-space control with LQR (Model 4) yielded a massive statistical improvement, achieving 80.0% path adherence and reducing the RMSE to 0.525 m. By penalizing state deviations and steering effort simultaneously, LQR smoothly tracked the racing line. However, because LQR lacks explicit constraint management, it is highly prone to overshoot during high-speed transients where the physical steering actuator limits are reached. This tracking limitation is directly compared in **Figure 4.3**. The top plot shows that the speed profile of the LQR (red line) is highly erratic, showcasing severe high-frequency chatter with spiky drops down to 2.0 m/s as the vehicle fights for traction. The bottom plot highlights how this velocity chatter directly translates to path deviations. The LQR cross-track error exhibits sharp, spiky peaks that surge up to 1.65 m. These surges correspond to high-speed corner entries where the unconstrained controller demanded steering commands that saturated the physical actuators.

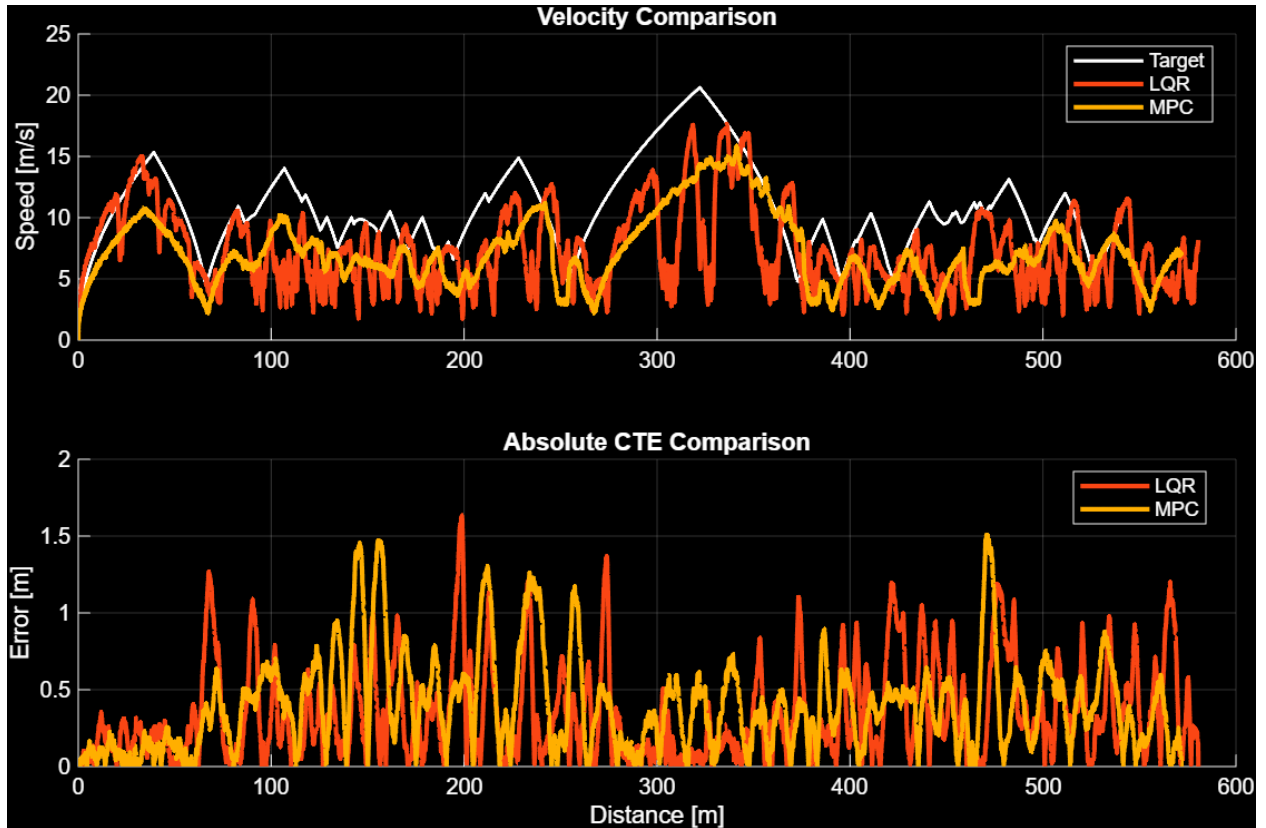


Figure 4.3: LQR vs MPC Performance (Velocity and CTE Comparison)

4.4.3 Predictive Superiority on a Clean Track (Model 5a)

To evaluate the unperturbed potential of the Explicit Model Predictive Controller, Model 5a was evaluated on a clean track with no obstacles. The system achieved a path adherence of 91.1%, an average velocity of 6.34 m/s, and a degraded RMSE of only 0.464 m. As illustrated in **Figure 4.4**, the red-orange trajectory line of the MPC is completely fused with the white reference path along the straightaways and the upper ridge. By predicting the track curvature over a 20-step horizon, the MPC optimizes the speed profile smoothly. Returning to the yellow curve of the velocity profile in **Figure 4.3**, it is evident that the MPC completely avoids the erratic, high-frequency speed chatter that degraded the LQR, ensuring that steering and throttle inputs remain within the stable linear region of tire slip.

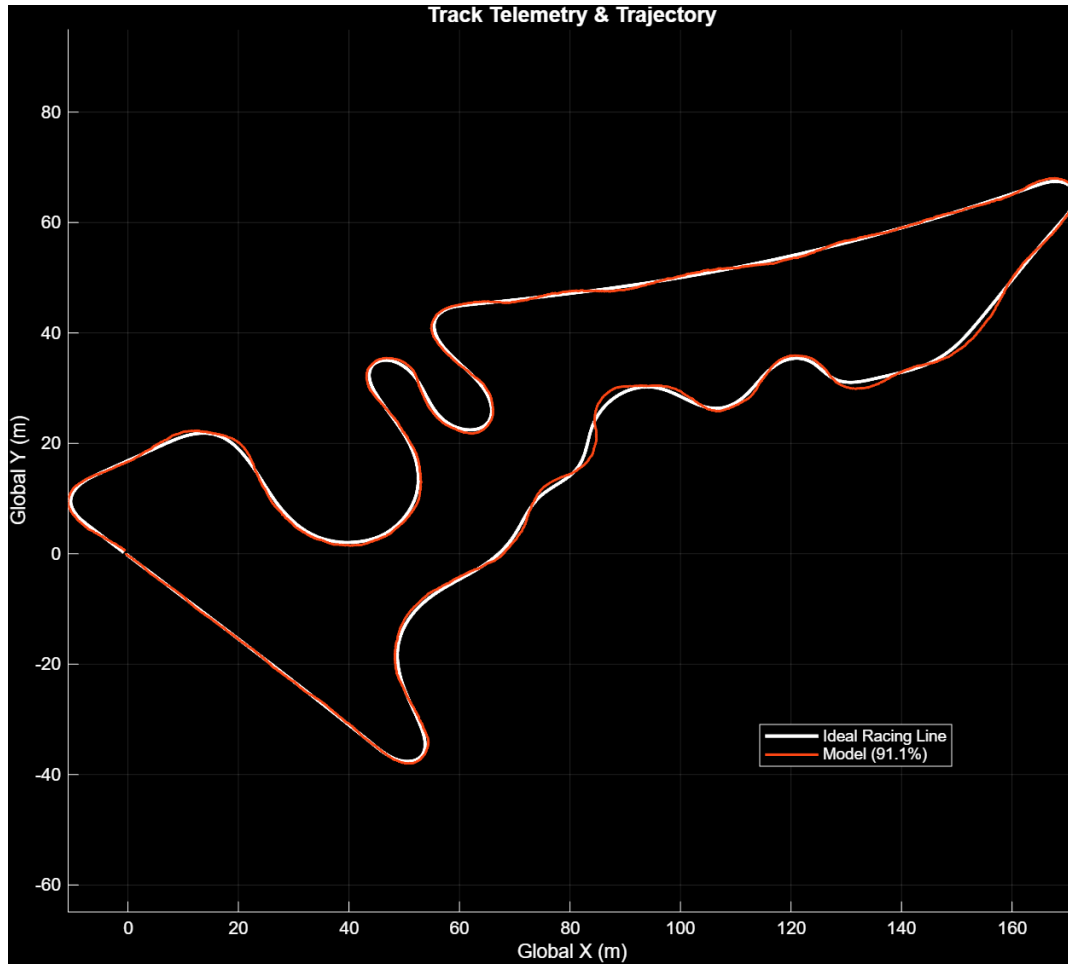


Figure 4.4: Model 5 Track Telemetry & Trajectory

4.4.4 High-Speed Obstacle Avoidance Case Study (Model 5b)

To test emergency maneuverability under extreme dynamics, Model 5b was subjected to a dynamic scenario where stalled obstacles were deterministically spawned directly on the ideal racing line. Under this active perturbation, the controller maintained dynamic equilibrium, achieving 88.8% path adherence, an average speed of 6.10 m/s, and an RMSE of 0.488 m. As illustrated in **Figure 4.5**, when the vehicle approaches the obstacles (indicated by red stars), the APF repulsion logic is activated, generating virtual lateral offsets (e_{apf}) that shift the track coordinates outward. The vehicle smoothly deflects from the ideal racing line to establish a safe passing corridor and immediately re-converges onto the track centerline once the obstacle is cleared.

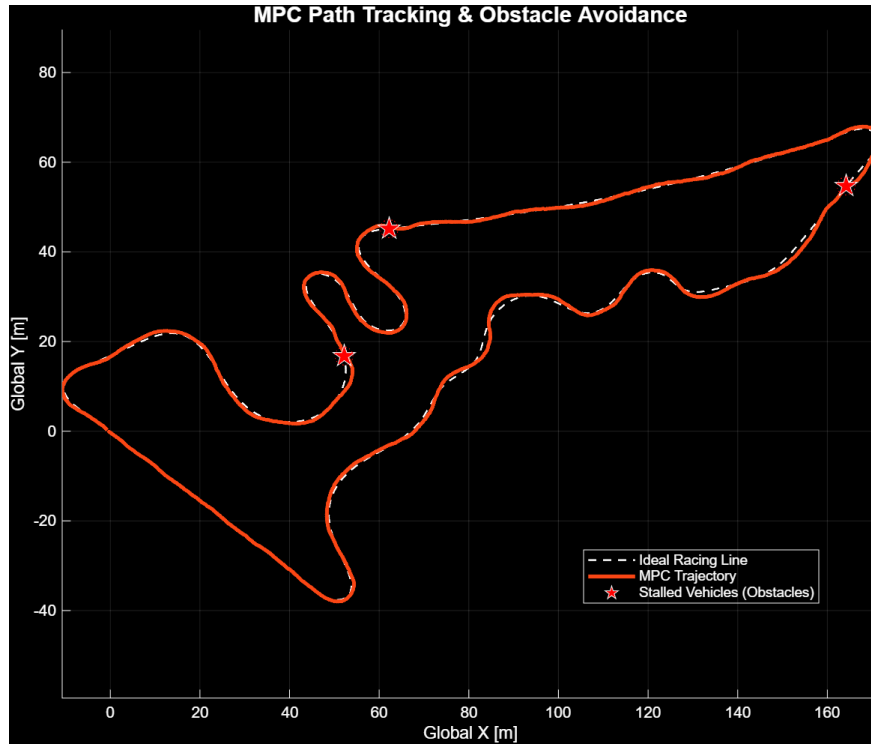


Figure 4.5: Global map trajectory tracking telemetry showing Model 5 multi-obstacle avoidance

Telemetry data from this run confirms the absolute necessity of the "Smart Throttle." When the APF injected a sudden lateral offset to dodge the obstacle, the required lateral acceleration spiked. To prevent violating the Kamm Friction Circle ($\mu = 0.6$), the

longitudinal governor detected the rising slip angles ($\alpha_f > 4.5^\circ$) and immediately throttled the velocity down toward the safety floor (5.0 m/s). Once the vehicle cleared the obstacle and the slip angles stabilized, the governor permitted re-acceleration, demonstrating a flawless synergy between predictive steering and active velocity management.

CHAPTER 5: DISCUSSION, CONCLUSION, AND SUGGESTION

This chapter interprets the quantitative simulation data presented in Chapter 4, providing a deep engineering analysis of why the control architectures succeeded or failed. Following this discussion, the chapter concludes the project outcomes, details the team's adherence to professional engineering ethics, and offers suggestions for future research and hardware deployment.

5.1 Discussion of Simulation Results

5.1.1 Master Performance Summary

Table 5.0 consolidates the final simulation metrics for all five control architectures at the maximum 20.0 m/s benchmark. This table clearly illustrates the evolutionary progression from catastrophic kinematic failure to stable predictive tracking.

Table 5.0: Master Simulation Performance Summary (20 m/s Benchmark)

Model Architecture	Path Adherence (%)	Average Speed (m/s)	RMSE Cross-Track (m)	Stability Status
Model 1: Pure Pursuit	22.2%	8.10	6.615	Terminal Understeer
Model 2: Stanley	34.0%	8.82	3.789	Terminal Understeer
Model 3: ActiveYaw PID	34.3%	8.47	3.352	Actuator Chatter
Model 4: LQR	80.0%	6.13	0.525	Transient Overshoot
Model 5a: MPC (Clean)	91.1%	6.34	0.464	Stable Limit Cycle
Model 5b: MPC (Obstacles)	88.8%	6.10	0.488	Stable Evasion

5.1.2 The Kinematic Disconnect: Why Baselines Failed

The catastrophic failure of the Pure Pursuit (6.615 m RMSE) and Stanley (3.789 m RMSE) controllers perfectly illustrates the danger of deploying kinematic planners in high-speed dynamic environments.

Both algorithms treat the vehicle as a geometric point mass, calculating steering angles based purely on positional error and heading mismatch. Because neither algorithm incorporates the 3DOF dynamic plant into its control law, they remain completely "blind" to the lateral forces (F_y) accumulating on the chassis.

As the vehicle enters a high-speed chicane, the geometric error grows, prompting the controllers to command increasingly aggressive steering angles (δ). This mathematically forces the front tire slip angle (α_f) well beyond the 5° Pacejka saturation limit. Once saturated, the tires lose lateral grip, rendering any further steering commands useless and trapping the vehicle in unrecoverable spin-outs. This mathematically proves that geometric steering is fundamentally inadequate at high speeds without integrated, dynamics-aware stability intervention.

5.1.3 Predictive Superiority and The Smart Throttle

The Explicit Model Predictive Controller (Model 5) achieved the definitive best performance due to two factors:

1. **The Receding Horizon:** Unlike reactive controllers, the MPC optimizes the trajectory over a 20-step prediction horizon ($p = 20$). By "looking ahead" at future path curvature, the MPC calculates a sequence of optimal steering adjustments before the vehicle reaches the corner.
2. **Constraint Enforcement:** The MPC mathematically enforces hard physical constraints (e.g., Steering $\leq \pm 30^\circ$, Slip Angle $\leq 5^\circ$) inside its optimization solver. It intentionally sacrifices minor cross-track precision to strictly enforce slip-angle constraints, preventing the spin-outs that plagued the baselines.

The true genius of this architecture is revealed in the vehicle's Phase Portrait, plotting chassis slip angle (β) against yaw rate (r), as presented in **Figure 5.1**. When the MPC is forced to execute a severe Artificial Potential Field (APF) lateral dodge, the physics undergo massive transient stress, captured as sweeping outer loops.

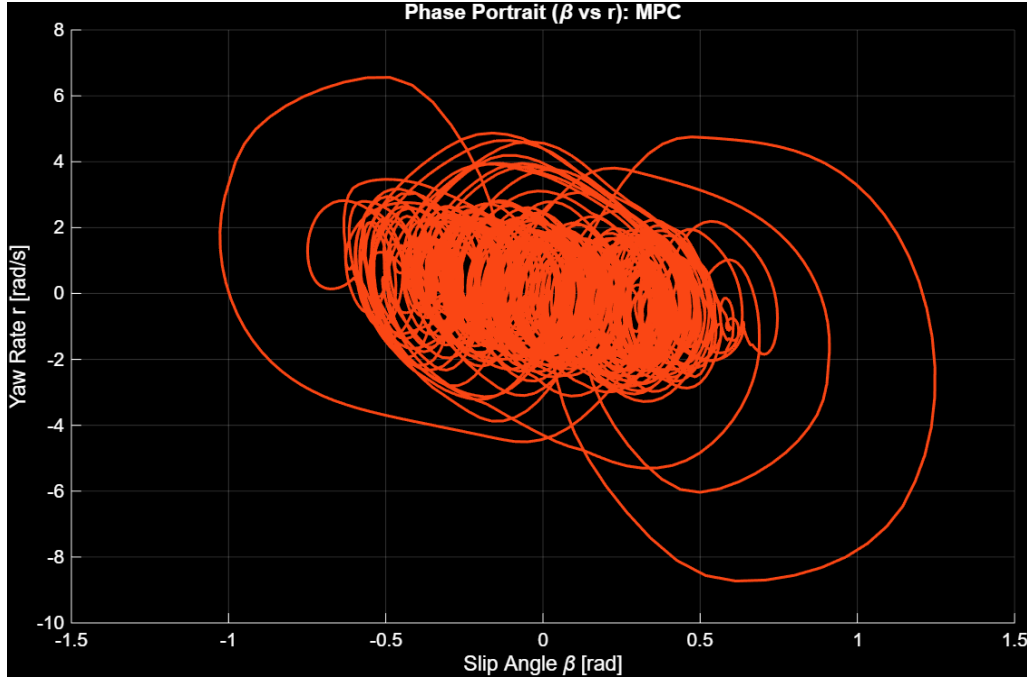


Figure 5.1: Phase Portrait (Slip Angle β vs Yaw Rate r)

If the controller relied on steering alone, these excursions would result in terminal spin-outs. Instead, the "Smart Throttle" longitudinal governor detects rising slip states and immediately derates the target velocity. By reducing speed, the system drastically shrinks the required lateral acceleration ($a_y = v_x^2/R$), safely dragging those massive transient loops back into the dense, stable central core of the phase portrait.

5.2 Conclusion

The primary objective of this project was to design, implement, and validate an Active Stability System capable of navigating complex racelines at high speeds. This was successfully achieved through a strategic Model-Based Design (MBD) methodology. By transitioning from a noise-limited hardware platform to a deterministic

3-Degree-of-Freedom (3DOF) digital twin, the project moved beyond basic obstacle avoidance into a master's-level study of vehicle handling at the friction limit.

The simulation experiments documented a clear evolution of control maturity. The geometric baselines mathematically proved that steering algorithms are physically "blind" to tire saturation. The Cascaded PID architecture enforced safety but suffered wide track errors. Ultimately, the Explicit Model Predictive Controller (MPC) coupled with the Smart Throttle friction interceptor achieved definitive success. By optimizing steering and velocity over a receding horizon, it maintained a sub-meter cross-track error (0.464 m) while strictly bounding tire slip angles below the 5° limit, even during APF obstacle evasion.

5.3 Engineering Professionalism and Teamwork

In accordance with KMUTT ENE ABET requirements, this section outlines the team's adherence to professional standards during the project lifecycle.

5.3.1 Team Brainstorming and The "Relay Marathon"

The successful completion of this project relied heavily on collaborative brainstorming and strategic decision-making.

1. **The Strategic MBD Pivot:** Early in Semester 1, the team faced persistent hardware failures (SSD corruption, SD card bit-rot, and ESP32 I/O latency). The team brainstormed the risks of continuing with hardware versus transitioning to simulation. **Resolution:** The team decided to pivot to a 100% Model-Based Design (MBD) environment, sacrificing physical hardware demonstration in favor of deep mathematical rigor and deterministic testing.
2. **The "Relay Marathon" Methodology:** Facing severe time constraints after the mid-semester pivot, the team brainstormed workflow optimizations. **Resolution:** The team adopted a "Pilot and Co-Pilot" pair-programming methodology on a single master workstation to prevent version control conflicts. One member focused on Simulink implementation while the other focused on real-time physics validation, ensuring rapid, error-free deployment.

5.3.2 Adherence to the IEEE Code of Ethics

This project strictly adheres to the IEEE Code of Ethics. Specific examples of compliance include:

1. **Honest Claims based on Data:** The third IEEE Code of Ethics states the obligation "to be honest and realistic in stating claims or estimates based on available data." Rather than manipulating physical hardware tests to appear successful under low-speed conditions, the team honestly documented the catastrophic failures of the baseline models. The final claims regarding the MPC's performance (0.464 m RMSE) are proven entirely through transparent, reproducible simulation data.
2. **Accepting Honest Criticism:** The fifth IEEE Code of Ethics states the obligation "to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors." During a critical consultation on March 9th, the project advisor critiqued the initial baseline methodology, noting that testing vehicles at a constant 20.0 m/s was a "self-created problem." The team accepted this valid criticism and overhauled the project to include Menger Curvature-based dynamic velocity profiling.

5.3.3 Awareness of Life-Long Learning

The rapid evolution of automation dictates that engineering education extends far beyond graduation. Classical PID control is a staple of undergraduate education, but this project revealed its severe limitations in handling multiple coupled states. Solving this required the team to independently learn Linear Quadratic Regulation (LQR) and Model Predictive Control (MPC).

A plausible situation requiring life-long learning is the transition of this project into the automotive industry. If hired to implement this system on a real autonomous vehicle, an engineer would need to independently learn strict automotive safety standards (such as ISO 26262), real-time C++ compilation protocols for embedded ECUs, and advanced sensor fusion techniques that were not covered in the scope of this undergraduate simulation.

5.3.4 Disclosure of AI Usage

In accordance with academic integrity guidelines, the team discloses the use of Artificial Intelligence tools (such as Gemini and Grammarly) during the compilation of this report. AI was strictly utilized as an advanced proofreading, formatting, and structuring tool to ensure clarity and professional phrasing. All engineering data, Simulink architectures, mathematical control laws, and technical conclusions are the original, authentic work of the project team.

5.4 Suggestions for Future Work

While the MPC architecture represents a significant technical achievement, several avenues remain for expanding the system's robustness:

1. **High-Fidelity 7DOF Modeling (CarSim):** The current 3DOF model operates within a single $X - Y$ plane, assuming a perfectly flat track. Future iterations should utilize 7-Degree-of-Freedom models (e.g., via CarSim integration) to analyze vertical load transfer during braking and the controller's robustness against variable track camber.
2. **Hardware Re-integration (C-Code Deployment):** With the control logic mathematically matured and the digital twin validated, the logical next step is hardware deployment. Future work will involve utilizing Simulink Coder to generate optimized C/C++ code from the MPC blocks, porting the algorithms back to the NVIDIA Jetson Nano.
3. **Extended Kalman Filter (EKF) Sensor Fusion:** To match the pristine state-estimation achieved in simulation on physical hardware, an EKF must be implemented to fuse the high-frequency IMU acceleration data with absolute velocity from wheel encoders, providing a drift-free state estimate resilient to real-world sensor noise.

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APPENDIX A

Project Source Code and MBD Repository

In strict adherence to modern software engineering standards and to ensure the total reproducibility of the Model-Based Design (MBD) findings presented in this study, the complete source code, simulation plants, and raw telemetry logs have been version-controlled and made publicly available.

GitHub Repository Link: <https://github.com/ajayi-ei/Active-Stability-MBD/>

A.1 Repository Architecture

The repository is structured to separate the non-linear physics plants, the execution scripts, and the exported visual analytics:

```
Active_Stability_MBD/
├── data/           # Simulation telemetry (.mat), 60+ Analytics Plots (.png), and Rendered Videos (.mp4)
├── models/        # Simulink 3DOF Non-Linear Physics Plants (.slx)
│   ├── M01_Baseline_PurePursuit.slx
│   ├── M02_Baseline_Stanley.slx
│   ├── M03_Cascaded_ActiveYaw.slx
│   ├── M04_LQR_SmartThrottle.slx
│   └── M05_MPC.slx
├── resources/     # Track matrices (Austin.csv) and Drone Tracker HUD assets (rc_car.png)
└── scripts/       # MBD Initialization, Execution, and Analytics code
    ├── car_init_M01... to M05...m # Plant and Controller Initialization Scripts
    ├── run_master_simulation.m    # Automated Batch Execution Pipeline
    ├── generate_analytics_suite.m # Solo/Pairwise A/B Plotting Engine
    ├── plot_results.m            # Live Workspace Telemetry Engine
    ├── animate_multi_car_race.m  # Synchronized 30FPS Cinematic Broadcast Engine
    └── animate_drone_view.m      # High-Fidelity Single-Car Optical Flow Tracker
```

A.2 Simulation Execution & Reproducibility Guide

To reproduce the exact empirical benchmark results and visual analytics discussed in Chapters 4 and 5, reviewers may execute the automated pipeline using the following steps:

1. **Environment Setup:** Clone the repository and open the root directory in MATLAB (R2025b or later). Add the `scripts/`, `models/`, and `resources/` directories to the active MATLAB path.
2. **Batch Execution:** Open and run `scripts/run_master_simulation.m`. This pipeline automatically initializes the 3DOF plant parameters, generates the spline-interpolated raceline, and logs telemetry across all five controller architectures.
3. **Analytics Generation:** Execute `scripts/generate_analytics_suite.m` to

automatically compile the high-contrast solo and pairwise A/B comparative plots (Trajectory, Velocity Profiling, Tire Slip α_f , Phase Portraits β vs r , and Actuator Chatter).

4. **Cinematic Telemetry:** Run `scripts/animate_multi_car_race.m` or `scripts/animate_drone_view.m` to render the dynamic broadcast videos showcasing the real-time simulation behavior.

A.3 Digital Twin Media & Simulation Videos

During the evaluation of high-speed dynamic stability, static trajectory plots only capture a fraction of the system's true behavior. To fully validate the dynamics of the "Kinematic Trap" and the subsequent recovery via predictive control, high-fidelity .mp4 video renders of the simulations have been generated and archived in the project repository.

1. The 5-Model Comparative Benchmark

This video overlays the real-time execution of all five control architectures operating simultaneously at the 20.0 m/s boundary limit. It provides visual proof of the catastrophic spin-outs suffered by the kinematic baselines (Models 1 & 2) and the actuator chatter of the PID controller, directly contrasted against the smooth limit-cycle tracking of the MPC.

Video Link: <https://youtu.be/RbYN7YcVuoo>

2. Model 5 (MPC) Solo Execution & Obstacle Evasion

This dedicated render isolates the Explicit Model Predictive Controller. It clearly demonstrates the real-time activation of the "Smart Throttle" (automatically braking before sharp chicanes) and the Artificial Potential Field (APF) smoothly injecting lateral offsets to evade stalled vehicles without destabilizing the chassis.

Video Link: <https://youtu.be/vVgv-OERw6k>

A.4 Authors

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